Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Agentic AI Systems · LLM Engineering · Python/AWS · Data Pipelines & Integrations · 9yr Production

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PROFESSIONAL SUMMARY

Senior software engineer with 9 years of production engineering depth, specializing in agentic AI systems, LLM engineering, and data pipeline development. I build AI-powered applications end-to-end: AgentProbe is a production LLM evaluation framework with a 5-metric eval engine, LLM-as-judge scoring, HITL escalation, hallucination detection, and CI-integrated regression gate -- the model evaluation and guardrails discipline this role requires. LeaseI is an 8-agent LangGraph agentic platform for document intelligence and information extraction with RAG retrieval, validation guardrails, and human-in-the-loop review -- the same agentic workflows and agentic frameworks CSC needs to embed into its cloud-native product stack. At Entrata, I owned production data pipelines processing millions of compliance-critical records annually, designed and maintained data models supporting analytics workloads, led AWS cloud migrations with 100% PHI compliance, and shipped 20+ full-stack modules across 120+ Agile sprints. I use AI-augmented development tools daily -- Claude Code (MCP-integrated) is my primary development environment. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

AI & Agentic Systems: LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, LLM APIs (Anthropic/Claude, OpenAI, Groq, Ollama), RAG Pipelines, Retrieval-Augmented Generation, Embeddings, Vector Databases (ChromaDB), Semantic Search, MCP Servers, Prompt Engineering, Output Validation, Guardrails, HITL (Human-in-the-Loop), LLM Evaluation, LLM-as-Judge, Hallucination Detection, Model Evaluation, Agent Observability & Tracing, Document Intelligence, OCR, Information Extraction

Languages & Frameworks: Python (9yr production), Node.js/JavaScript, TypeScript, C#/.NET (1.5yr), Java, SQL/PL-SQL, PHP, R, Bash, HTML/CSS; FastAPI, Flask, REST APIs, React.js; data modeling, data migrations, ETL/ELT pipelines, middleware integration

Cloud, Infrastructure & Practices: AWS (S3, EC2, RDS, Glue), cloud native application development, Cloudflare (learning), Docker, Kubernetes, CI/CD (GitHub Actions), Git, PostgreSQL, MySQL, Oracle, NoSQL; SDLC, Agile/Scrum, Code Reviews, Security & Compliance, Performance Optimization; mentoring junior engineers; AI-augmented development -- Claude Code (MCP) daily

EXPERIENCE

Magizhs Tech Solution

May 2025 -- Aug 2025

AI/ML Engineer -- Agentic Systems & Workflow Automation

- Built an agentic mission-planning system as a LangGraph stateful graph with conditional routing and guardrails: parse → plan → retrieve → validate → export -- demonstrating the end-to-end agentic workflow automation pattern CSC's Innovation team requires.
- Implemented RAG retrieval over a regulatory corpus (FAA Part 107) with cited sources; engineered validation guardrails enforcing compliance constraints before export -- blocking non-compliant outputs with grounded rejection reasoning.

George Mason University

Aug 2024 -- May 2026

Research Associate & Graduate Teaching Assistant

- Built multi-source ML prediction pipelines (Python, R) across 190+ countries (R^2=0.977); applied Isolation Forest to automate data integrity checks across 3 research pipelines; designed Tableau/ggplot2 dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and data engineering.

Entrata -- Multifamily Real Estate SaaS (PropTech)

Sep 2018 -- Aug 2024

Senior Software Engineer -- Data Engineering, Backend Systems & Cloud Infrastructure · Lehi, UT (Remote)

- **Production data pipelines and data model ownership.** Sole owner of Entrata's annual HUD income-limit pipeline for 7 years: federal regulatory data transformed and mapped to ~180,000 domain records powering eligibility and rent calculations for 100K+ housing units. Defined data transformation rules, validation logic, data migrations, and reconciliation workflows. Zero data integrity violations across the full ownership period.
- **Scalable backend services and cloud infrastructure.** RabbitMQ-based async processing with idempotency guarantees; optimized 200+ SQL queries (30% performance gain) through execution plan analysis, indexing, and stored procedure refactoring; Docker microservices redesign (50% scalability improvement, 30% deployment time reduction); AWS S3 migration with Libsodium-encrypted ETL pipelines (100% PHI compliance, zero security incidents); 99.5% uptime.
- **Integration architecture and automation.** Built and maintained integrations between internal CRM systems and external vendor applications via middleware and API contracts; managed data import pipelines from multiple third-party sources; eliminated 80% of manual operational workload through Python/Bash automation pipelines across the SDLC.
- **Technical leadership.** Mentored and leveled up 5+ engineers in AI fluency and cloud-native practices; conducted 300+ peer code reviews; built internal debugging tools reducing debugging time 40%; championed CI/CD, observability, and data models standards; maintained bug count below 10 for 6+ consecutive months.

Entrata India -- Pune

Jan 2017 -- Sep 2018

Associate Software Engineer -- Python · Java · C# · PostgreSQL

- Built compliance data pipelines (income-limit import, 1.6M HUD/Tax Credit records); AWS S3 cloud migration; database schema redesign with post-deployment stabilization; full-stack modules in PHP, Python, Java, and C#.

PROJECTS

AgentProbe: Production LLM Evaluation Framework

LLM Evaluation · HITL · Guardrails · Model Evaluation · CI/CD · Observability

Python · LangGraph · FastAPI · ChromaDB · React.js · TypeScript · GitHub Actions | github.com/venkatalakshmiparimala/AgentProbe

Production AI evaluation and observability framework: 5-metric eval engine (correctness, groundedness, hallucination rate, latency, token efficiency), LLM-as-judge scoring with multi-trial pass@k harness, ChromaDB vector evaluation, CI-integrated regression gate blocking quality degradation, structured agent tracing for failure analysis, multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama), HITL escalation for low-confidence outputs. Establishes the model evaluation practices, prompt engineering standards, and guardrails patterns CSC's AI governance framework requires.

Leasel: Agentic AI Platform -- Document Intelligence & RAG

LangGraph · Agentic AI · RAG · HITL · Guardrails · LLM Orchestration

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · OCR · SQLite · GitHub Actions | github.com/venkatalakshmiparimala/Lease_complaince_rag

End-to-end agentic AI platform demonstrating autonomous and semi-autonomous agent design: 8-agent LangGraph pipeline with conditional routing, OCR-fallback document ingestion, TF-IDF RAG retrieval with citation grounding, HITL escalation gate on regulated decisions, and validation guardrails enforcing compliance constraints. 93.6% precision, 87.9% F1 across 1,240 test cases (5 US states + federal regulatory corpus). Directly demonstrates the agentic workflow orchestration, RAG integration, and responsible AI guardrails CSC's platform needs.

AI on FHIR: Secure Health Data Platform -- HIPAA & RBAC

Flask · REST API · HIPAA · RBAC · NLP · Security & Compliance

Python · Flask · React · Docker · FHIR API · spaCy · BERT | github.com/venkatalakshmiparimala/AI_on_FHIR

HIPAA-compliant AI platform integrating NLP (BERT, spaCy) with FHIR-standard health data APIs; RBAC security controls and compliance governance for AI-generated outputs. Demonstrates the security and compliance framework for AI systems this role requires.

EDUCATION

M.S. Data Analytics Engineering -- George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering -- JNTU Anantapur — 2013--2017